LAB 03

SOFTWARE: The image was converted to grayscale, and the matrix of the grayscale image was obtained in a .txt format using Python.

Original Image

Grayscale Image

Convert image to grayscale matrix

```
import cv2
import numpy as np

# Upload the image file to Google Colab's runtime environment

# Load the uploaded image
image = cv2.imread('/content/HSCD_Lab3.jpg')

# Convert the image to grayscale
gray_image = cv2.cvtColor(image, cv2.COLOR_BGR2GRAY)

# Convert the grayscale image to an integer matrix
integer_matrix = gray_image.astype(int)

# Save the integer matrix to a text file
np.savetxt('/content/grayscale_matrix.txt', integer_matrix, fmt='%d')

print("Integer matrix saved to grayscale_matrix.txt")
```

```
Integer matrix saved to grayscale_matrix.txt
```

Displaying the grayscale image

```
import numpy as np
import cv2
from google.colab.patches import cv2_imshow

# Read the integer matrix from the text file
integer_matrix = np.loadtxt('/content/grayscale_matrix.txt', dtype=int)

# Generate the grayscale image from the integer matrix
gray_image = integer_matrix.astype(np.uint8)
```

```
# Display the grayscale image
cv2_imshow(gray_image)
```

HARDWARE: The dimension of the grayscale matrix obtained was 100x88, which was then read in a Verilog code. The Gx and Gy matrices' dimensions were obtained from the formula n-k+1, where n and k are the dimensions of the grayscale matrix and Sobel Kernels respectively. On performing the convolution for Gx and Gy and applying the Sobel Filter, we generated the final matrix.

Design:

design.sv | grayscale_matrix.txt

```
module sobel_filter();
    integer fd,code,i,j,k,l;
    reg [31:0] MEM[0:99][0:87];
    reg [31:0] G1[0:2][0:2];
    reg [31:0] G2[0:2][0:2];
    reg signed[31:0] Gx[0:97][0:85];
    reg signed[31:0] Gy[0:97][0:85];
    reg signed[31:0] G[0:97][0:85];
    reg [31:0] s;
    initial
        begin
            //G1
            G1[0][0] = 1;
            G1[0][1] = 0;
            G1[0][2] = -1;

            G1[1][0] = 2;
            G1[1][1] = 0;
            G1[1][2] = -2;

            G1[2][0] = 1;
            G1[2][1] = 0;
            G1[2][2] = -1;

            //G2
            G2[0][0] = 1;
            G2[0][1] = 2;
            G2[0][2] = 1;

            G2[1][0] = 0;
            G2[1][1] = 0;
            G2[1][2] = 0;

            G2[2][0] = -1;
            G2[2][1] = -2;
            G2[2][2] = -1;

            //reading the grayscale matrix text
             fd = $fopen("grayscale_matrix.txt", "r");
                for(i=0; i<100; i++)
                  begin
                    for(j=0; j<88; j++)
                      begin
                        code = $fscanf(fd, "%d", s);
```

design.sv | grayscale_matrix.txt

```
                td = $topen("grayscale_matrix.txt", "r");
                for(i=0; i<100; i++)
                  begin
                    for(j=0; j<88; j++)
                      begin
                        code = $fscanf(fd, "%d", s);
                        MEM[i][j] = s;
                      end
                  end

                //Convolution for Gx & Gy
                for(i=0; i<98; i=i+1)
                      begin
                        for(j=0; j<86; j=j+1)
                          begin
                              Gx[i][j] = 0;
                              Gy[i][j] = 0;
                              for(k=0; k<3; k=k+1)
                                  begin
                                  for(l=0; l<3; l=l+1)
                                      begin
                                        Gx[i][j] = Gx[i][j] + (MEM[k+i][l+j]*G1[k][l]);
                                        Gy[i][j] = Gy[i][j] + (MEM[k+i][l+j]*G2[k][l]);
                                      end
                                  end
                          end
                      end

                //Sobel Matrix G
                for(i=0; i<98; i=i+1)
                      begin
                        for(j=0; j<86; j=j+1)
                              G[i][j] = $sqrt((Gx[i][j]*Gx[i][j]) + (Gy[i][j]*Gy[i][j]));
                      end

                //Display Sobel Matrix
                for(i=0; i<98; i=i+1)
                      begin
                        for(j=0; j<86; j=j+1)
                          $write("%d ",G[i][j]);
                          $display(" ");
                      end
                      $finish;
                  end
endmodule
```

SOFTWARE:

The filtered matrix was stored in a .txt file which was converted to an image using Python. The image on applying Sobel filter has been converted to an image that emphasizes edges.